JENNIFER ZHU

RESEARCH DATA ANALYST | BIOSTATISTICS

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PROFILE

Research data analyst with graduate training in Health Data Science and Statistics, experienced in longitudinal patient-level data, oncology survival analysis, multiple imputation, Bayesian modeling, reproducible analysis and manuscript figures.

RESEARCH EXPERIENCE

Graduate Student Researcher | UCSF, Lupo Lab | June 2025–Present

- Analyze longitudinal multiparametric MRI and clinical outcomes across 95 patients with newly diagnosed glioblastoma from three research cohorts, covering pre- and post-radiotherapy, pre-progression, and progression scans.
- Perform Kaplan–Meier, multivariable Cox regression, linear regression, model diagnostics, and false discovery rate adjustment to assess imaging–outcome associations.
- Design and adapt ROI-based quantitative imaging feature workflows for contrast-enhancing and non-enhancing lesions, including distributional statistics, tumor-volume changes, and radiopathomic cellularity and tumor-probability maps.
- Create MRI overlays, survival plots, manuscript figures, and scientific summaries for multidisciplinary collaborators; first-author manuscript in preparation.

SELECTED PROJECTS

METABRIC — Clinical Survival Analysis

- Analyzed hormone therapy and recurrence-free survival in an ER-positive breast cancer cohort using Kaplan–Meier, multiple imputation, and adjusted/stratified Cox regression.
- Used Schoenfeld residuals for proportional-hazards diagnostics and Rubin's rules for pooling estimates.

I-SPY2 — Bayesian Modeling of Treatment Response

- Fit Bayesian logistic regression models of pathological complete response using baseline MRI tumor diameter, sphericity and HR/HER2 status.
- Compared three models using leave-one-out cross-validation; assessed posterior distributions, MCMC convergence and prior sensitivity on 384 complete cases.

BloodMNIST — Biomedical Computer Vision

- Collaborative project comparing a PyTorch convolutional neural network with a random-forest baseline on eight classes of microscopy images.
- CNN achieved 90.76% held-out test accuracy; evaluated precision, recall, F1 and ROC-AUC across classes.

EDUCATION

M.S., Health Data Science | University of California, San Francisco | June 2026

B.S., Statistics | San Francisco State University | December 2022

TECHNICAL SKILLS

Clinical Statistics: Kaplan–Meier; Cox PH; stratified Cox; Schoenfeld residual diagnostics; multivariable regression; MICE; Rubin's rules; FDR; survival outcomes

Bayesian & Statistical Learning: Bayesian logistic regression; prior specification; MCMC; posterior predictive checks; LOO-CV; logistic/linear regression; PCA; QDA

Tools & Reporting: R; Python; SQL; SAS; MATLAB; tidyverse; ggplot2; pandas; Matplotlib; Jupyter; R Markdown; Tableau; Power BI; Git

RESEARCH OUTPUT

First-author manuscript in preparation — longitudinal radiopathomic MRI features and clinical outcomes in glioblastoma (UCSF Lupo Lab).